

23ECE286 Circuits and Communication Laboratory

**Design and implement a circuit to create a hysteresis window
between 1V and 3V for a noisy input signal.**

Term Project Report

Submitted by

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1. Introduction

The objective is to design and implement an Inverting Schmitt Trigger circuit that introduces a hysteresis window between 1V and 3V. The circuit should prevent rapid switching of the output caused by noise and ensure stable operation

In many electronic circuits, unwanted noise in the input signal can cause the output to switch rapidly between HIGH and LOW, leading to unstable behaviour. To prevent this, a hysteresis window can be added using an Inverting Schmitt Trigger.

In an **Inverting Schmitt Trigger**, the input signal is applied to the inverting input of a comparator or operational amplifier. The output behaviour is as follows:

- **Upper Threshold (V_{UT}) = 3V:**

When the input voltage exceeds 3V, the output switches to **LOW**.

- **Lower Threshold (V_{LT}) = 1V:**

When the input voltage drops below 1V, the output switches to **HIGH**.

This approach ensures that the circuit ignores small noise variations between the thresholds, providing a stable and reliable output

2. Objectives:

The main objective is to design and build an **Inverting Schmitt Trigger circuit** with a 1V to 3V hysteresis window. The specific goals include:

1. Circuit Design:

- Design an inverting Schmitt Trigger using an operational amplifier or comparator.
- Calculate the resistor values to achieve upper and lower thresholds at 3V and 1V.

2. Simulation and Analysis:

- Simulate the designed circuit using LTSpice.
- Verify that the circuit responds correctly to input signals with noise.

3. Hardware Implementation:

- Assemble the circuit on a breadboard or PCB.
- Test the circuit with different input signals (such as square wave, sine wave, and noisy signals).

4. Performance Evaluation:

- Measure the upper and lower threshold voltages.
- Analyze the circuit's response to noisy inputs and ensure it performs as expected.

3. Methodology

We started the project by researching different circuit configurations to introduce a hysteresis window for noisy input signals. After exploring various approaches, we found that an Inverting Schmitt Trigger was the most effective option for maintaining stable output.

The design phase began with calculating the resistor values to set the upper and lower thresholds at 3V and 1V using standard voltage divider formulas. A suitable reference voltage was selected to ensure that the circuit behaved correctly.

To validate the design, we used LTSpice to simulate the circuit and analyze its performance with different types of input signals, including noisy signals. The simulation results showed that the output correctly switched to LOW when the input crossed 3V and switched to HIGH when the input fell below 1V. After successful simulation, we used Eagle to design the PCB layout for hardware implementation.

4. Circuit Design and Components:

The circuit is designed to implement an Inverting Schmitt Trigger that creates a hysteresis window between 1V and 3V to stabilize a noisy input signal. An LM741 operational amplifier (U2) is configured as an inverting comparator with positive feedback provided by R₁ (130k Ω) and R₂ (10k Ω), which dynamically adjust the threshold voltages. The input signal, a 10V amplitude sinusoid at 100 Hz, is applied through R₁ (10k Ω) to the inverting terminal,

with additional white noise from B1 simulating real-world noise. A voltage divider formed by R4 (6.1k Ω) and R5 (1k Ω) sets the reference voltage ($V_{ref} \approx 2.11V$) for the non-inverting terminal

Calculation:

Given,

$$V_{UT} = 3V \text{ and } V_{LT} = 1V$$

$$\pm V_{CC} = 15V$$

Assuming,

$$\pm V_{SAT} = 0.9 \times \pm V_{CC}$$

$$\pm V_{SAT} = 0.9 \times \pm 15V$$

$$\pm V_{SAT} = \pm 13.5V$$

Hysteresis voltage V_H for inverting Schmitt trigger is given by,

$$V_H = V_{UT} - V_{LT}$$

$$V_H = 3 - 1$$

$$V_H = 2V$$

Also,

$$V_H = 2 \left(\frac{R_1}{R_1 + R_2} \right) * V_{SAT}$$

$$\frac{1}{V_{SAT}} = \left(\frac{R_1}{R_1 + R_2} \right)$$

$$\frac{1}{13.5} = \left(\frac{R_1}{R_1 + R_2} \right)$$

$$0.074 = \left(\frac{R_1}{R_1 + R_2} \right)$$

$$0.074 R_1 + 0.074 R_2 = R_1$$

$$0.074 R_2 = R_1 - 0.074 R_1$$

$$0.074 R_2 = 0.926 R_1$$

$$R_2 = 12.513 R_1$$

$$\text{Let } R_1 = 10\text{k}\Omega$$

$$\therefore R_2 = 12.513 * 10 * 10^3$$

$$R_2 = 125\text{k}\Omega \approx 130\text{k}\Omega \text{ (E24 Series)}$$

Centre Voltage of hysteresis is given by,

$$V_{CTR} = \frac{V_{UT} + V_{LT}}{2}$$

$$V_{CTR} = \frac{3 + 1}{2} = \frac{4}{2} = 2V$$

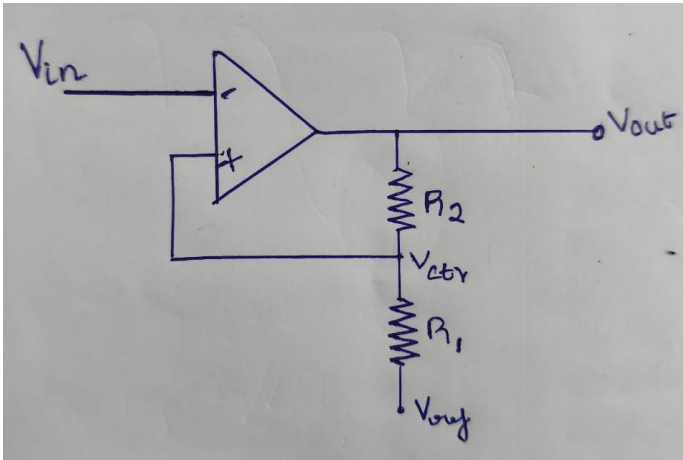


Fig. 1. +ve Feedback Circuit

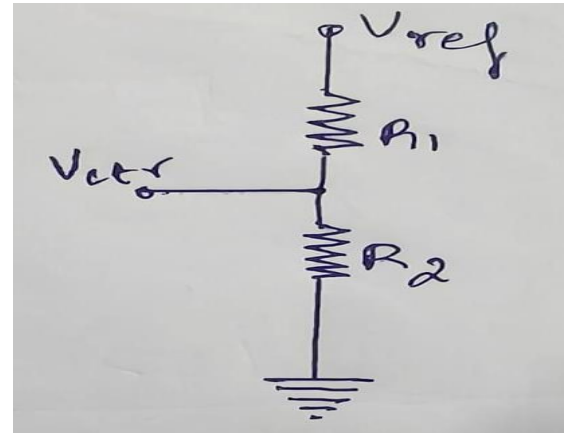


Fig. 2. Voltage Divider Circuit

In Fig.2, using Superposition Principle, keeping V_{REF} is active (i.e.) V_{OUT} is deactivated,

$$V_{CTR} = \left(\frac{R_2}{R_1 + R_2} \right) * V_{REF}$$

$$2 = \left(\frac{R_2}{R_1 + R_2} \right) * V_{REF}$$

$$V_{REF} = 2 \left(\frac{R_1 + R_2}{R_2} \right)$$

$$= 2 \left(\frac{10K + 125K}{125K} \right)$$

$$= 2 \left(\frac{135K}{125K} \right) = 2(1.08)$$

$$V_{REF} = 2.16V$$

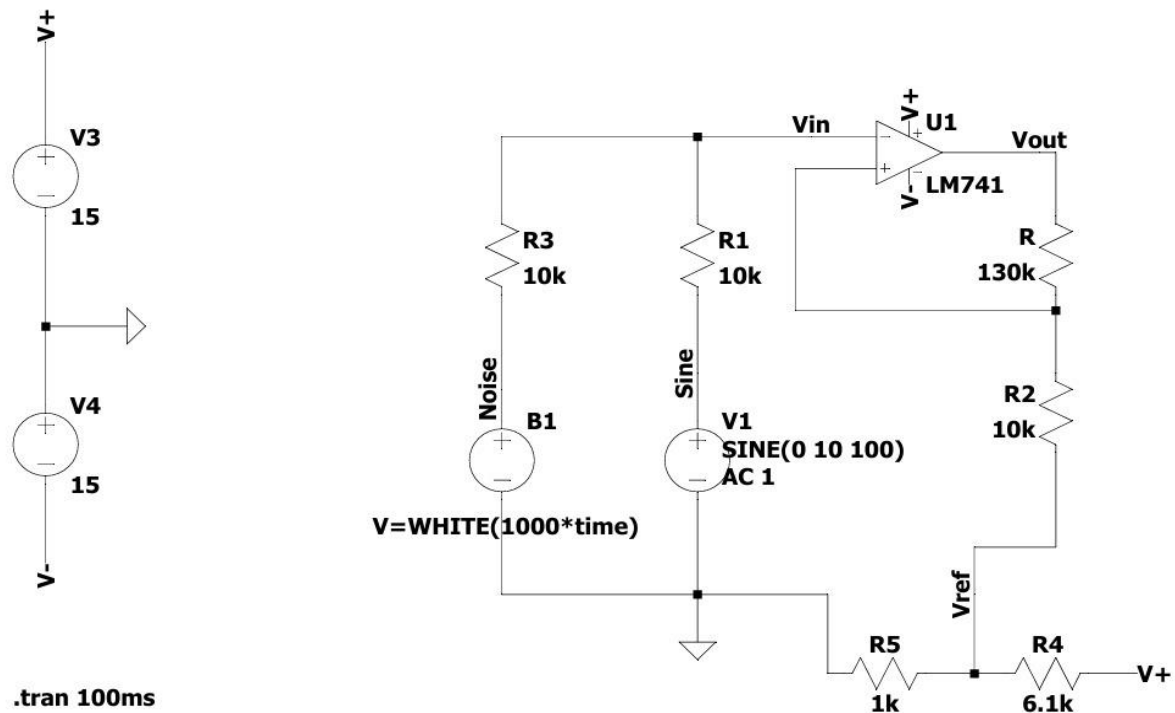


Fig. 3. Designed Circuit

Components Required:

S.No	Name of The Component	Required Quantity
1.	DC Power Supply (0-200V)	1
2.	Op-Amp (LM741)	1
3.	Resistors (1kΩ, 6.1kΩ, 10kΩ, 130kΩ)	(1,1,3,1) each
4.	Function Generator	1
5.	Digital Storage Oscilloscope	1
6.	Breadboard	1
7.	Connecting Wires	As Required
8.	Probes	As Required

DC Power Supply (0-200V):

- Provides the required $\pm 15\text{V}$ supply to power the op-amp.
- Ensures stable operation of the circuit by maintaining a constant voltage.

Op-Amp (LM741):

- Acts as a comparator in the **Inverting Schmitt Trigger** configuration.
- Compares the input voltage with the dynamically changing threshold and switches the output accordingly.

Resistors (1k Ω , 6.1k Ω , 10k Ω , 130k Ω):

- Set the reference voltage and control the upper and lower threshold voltages.
- Provide positive feedback to introduce hysteresis and stabilize the circuit.

Function Generator:

- Generates the sinusoidal input signal with an amplitude of 10V and frequency of 100 Hz.
- Allows testing of the Schmitt Trigger's response to noisy input signals.

Digital Storage Oscilloscope (DSO):

- Monitors and visualizes the output waveform to verify circuit performance.
- Measures switching thresholds and ensures the correct operation of the circuit.

Breadboard:

- Provides a platform for assembling and testing the circuit.
- Facilitates easy connection and modification of circuit components.

Connecting Wires:

- Establish connections between the components.
- Ensure smooth signal transmission without interference.

Probes:

- Used to connect the oscilloscope and measure signals at various points.
- Helps in accurate visualization and analysis of circuit performance.

5. Implementation:

The circuit was built by first assembling the components on a breadboard. A DC power supply was configured to provide $\pm 15\text{V}$ to power the LM741 op-amp. The input signal, a 10V amplitude sinusoid at 100 Hz, was generated using a function generator and applied through a $10\text{k}\Omega$ resistor (R_1) to the inverting terminal of the op-amp. Positive feedback was provided by the $130\text{k}\Omega$ resistor (R) and the $10\text{k}\Omega$ resistor (R_2) to establish the required hysteresis window. A voltage divider formed by R_4 ($6.1\text{k}\Omega$) and R_5 ($1\text{k}\Omega$) generated the reference voltage ($V_{\text{ref}} \approx 2.11\text{V}$) for the non-inverting terminal.

The circuit was simulated in LTSpice to verify correct operation before proceeding to hardware implementation. Once the design was confirmed, the hardware performance was validated using a digital storage oscilloscope (DSO).

Challenge during Implementation:

One major challenge encountered was the inability to generate white noise in the hardware implementation, which was successfully simulated in LTSpice. As a result, we were unable to precisely replicate real-world noise conditions during hardware testing. Another significant challenge was matching the exact resistor values calculated during design with the available resistor values in the E24 series (standard resistor values). Since exact values were not available, we had to use the closest available resistors, which required recalculating the threshold voltages to ensure the desired hysteresis window was maintained. Despite these challenges, after careful resistor selection and fine-tuning, the circuit was able to meet the required performance criteria.

6. Results:

Simulated:

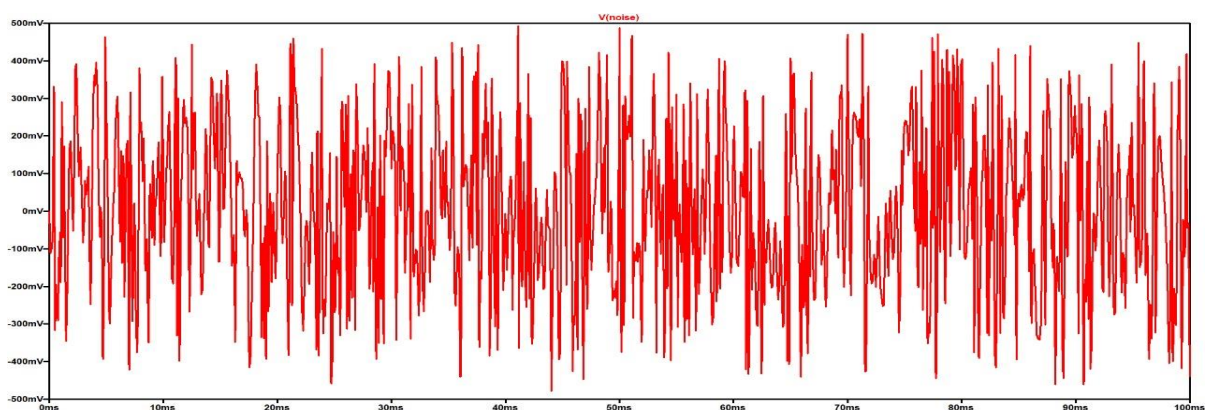


Fig. 4. Graph of Noise

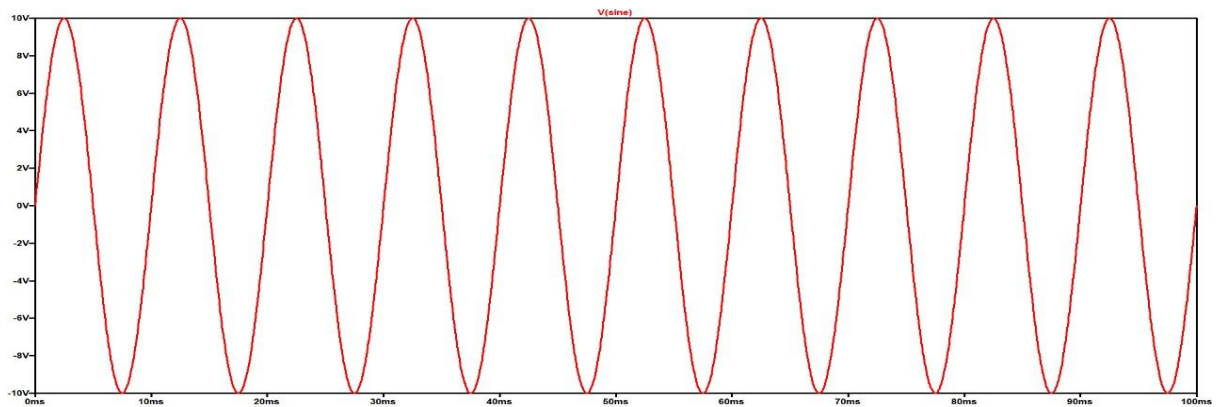


Fig. 5. Graph of Input Signal

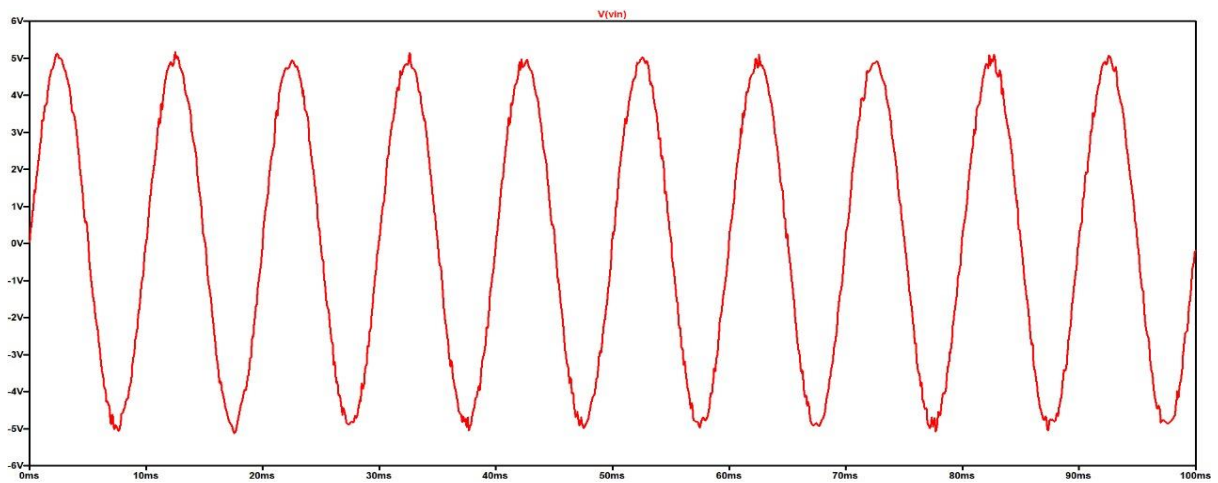


Fig. 6. Graph of Noisy Input Signal

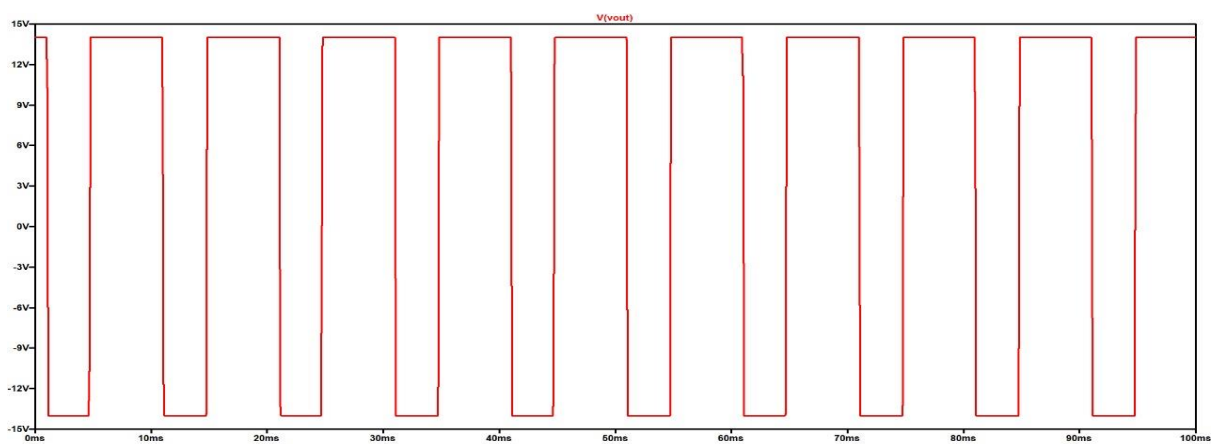


Fig. 7. Graph of Output Signal

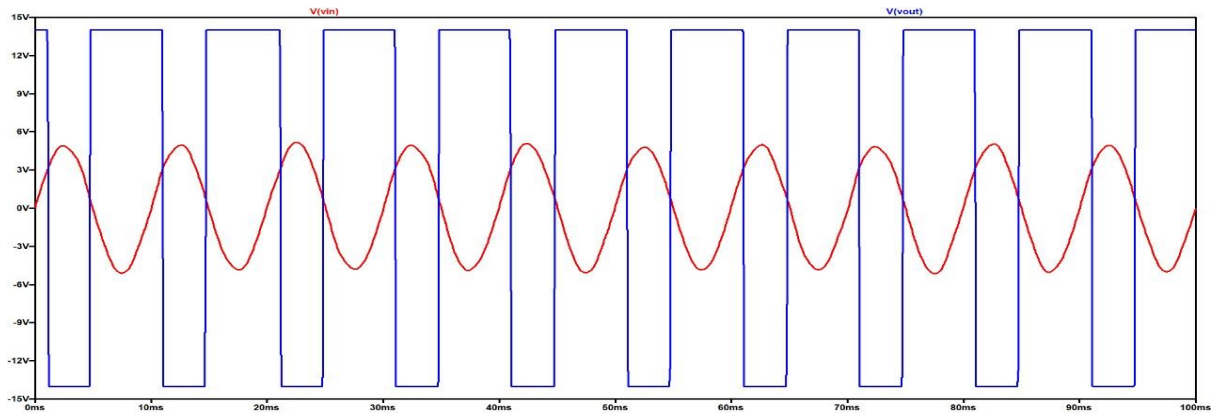


Fig. 5. Graph of Output Signal with Noisy Input Signal

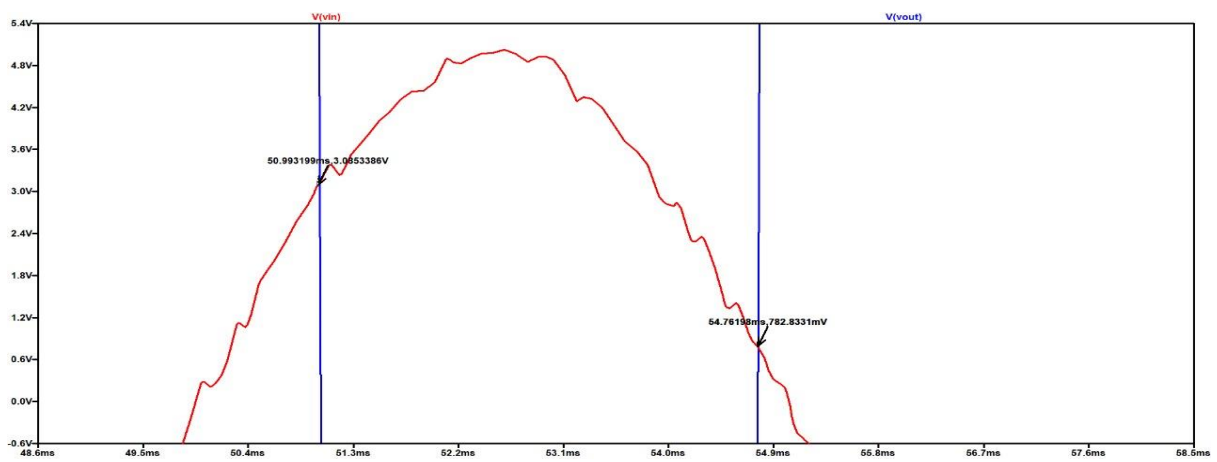


Fig. 6. Graph of Threshold Points

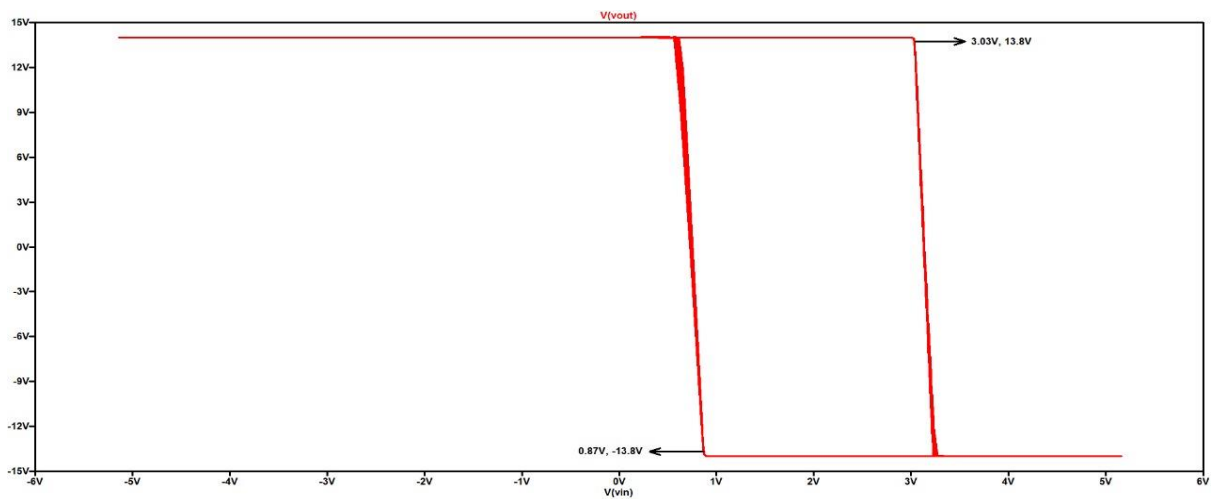


Fig. 7. Graph of Hysteresis Window

Implemented:

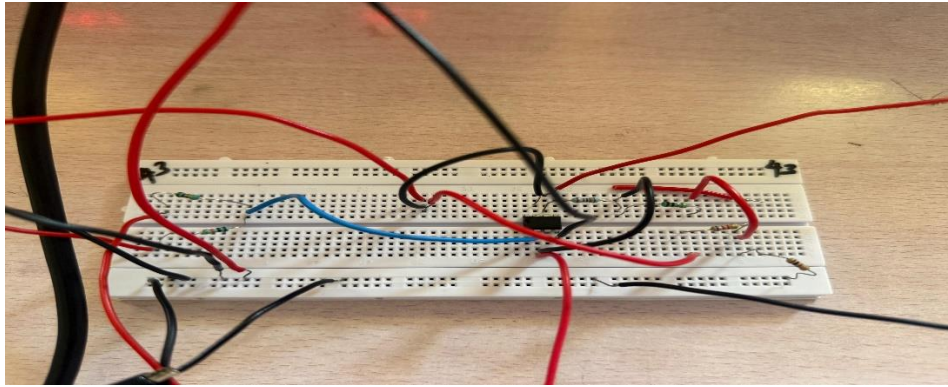


Fig. 8. Circuit Implemented

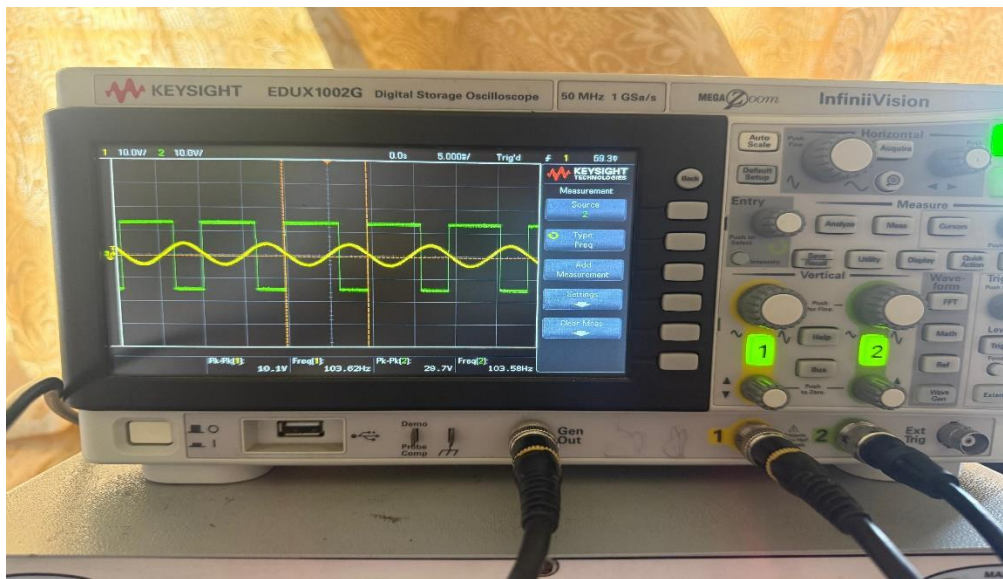


Fig. 9. Input and Output Waveform in DSO

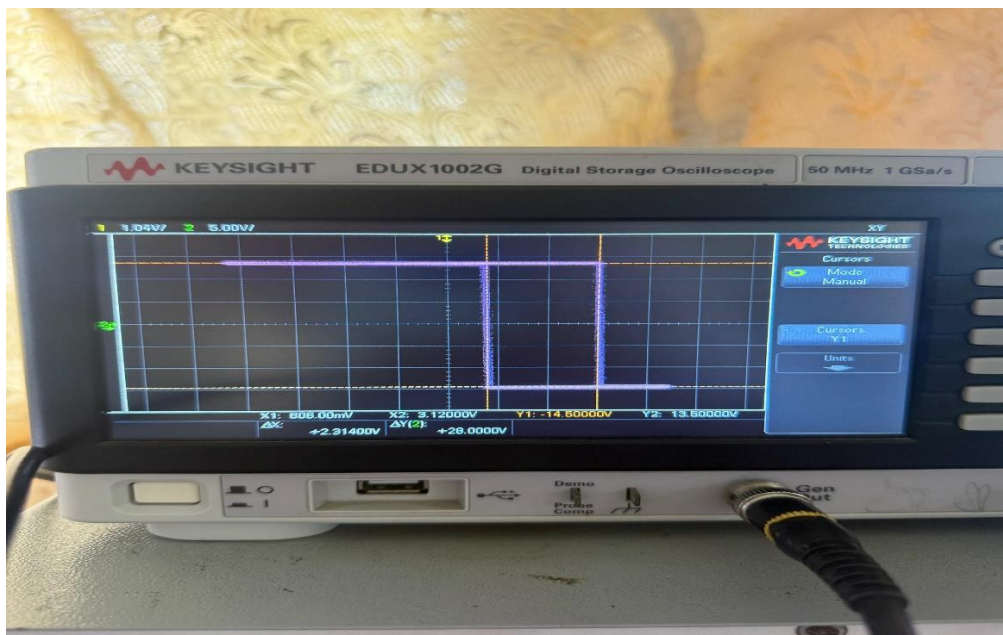


Fig. 10. Hysteresis Window between 1V and 3V in DSO

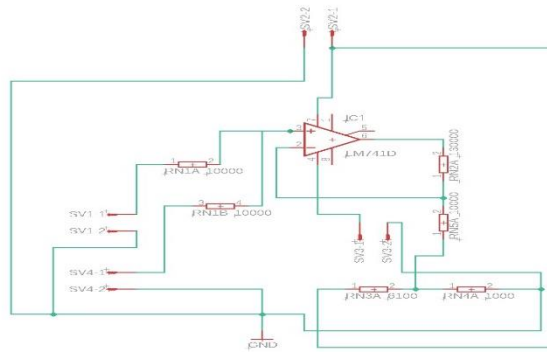


Fig. 11. PCB Schematic Diagram

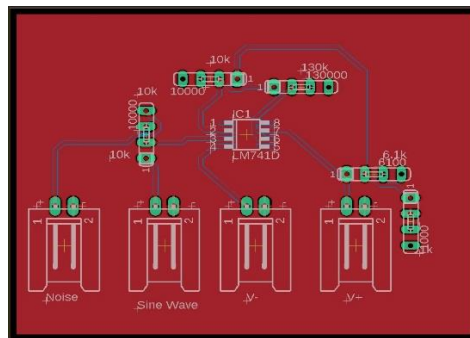


Fig. 12. PCB Layout

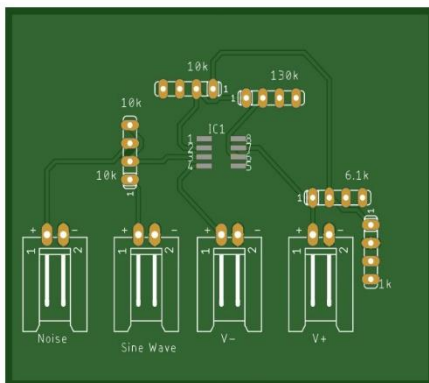


Fig. 13. PCB Printable Form (Front)

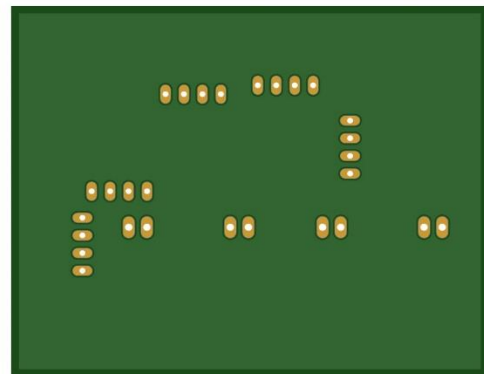


Fig. 14. PCB Printable Form (Backside)

7. Discussion:

- The circuit successfully implemented an Inverting Schmitt Trigger, introducing a hysteresis window between 1V and 3V to stabilize noisy input signals. When the input voltage exceeded 3V, the output switched to LOW, and when it dropped below 1V, the output switched to HIGH. The output waveform, observed using a Digital

Storage Oscilloscope (DSO), confirmed that the circuit maintained the desired hysteresis window, effectively filtering noise and preventing rapid oscillations.

- Minor deviations were observed between the simulation and hardware results. One key deviation was the absence of simulated white noise during hardware testing, as we couldn't generate a similar noise signal in the hardware setup. Another deviation occurred due to the unavailability of exact resistor values in the E24 series, requiring the use of the closest available values. These variations caused slight shifts in the threshold voltages, which were adjusted to maintain the desired hysteresis.
- The project successfully achieved its objective of designing and implementing a circuit to create a stable hysteresis window between 1V and 3V. Despite minor deviations, the circuit effectively stabilized the output and prevented unnecessary switching due to noise. The use of an Inverting Schmitt Trigger proved to be a reliable solution, and both simulation and hardware testing confirmed that the circuit met the desired specifications.

8. Challenges and Solutions:

Challenges:

One of the major challenges faced during the project was the absence of a third voltage source in the lab setup. The initial design required three voltage sources, but since only two sources were available, we had to create the third voltage using a voltage divider. By carefully selecting resistor values, we generated the required reference voltage for the non-inverting terminal of the op-amp, ensuring that the circuit operated as intended.

Another challenge was the inability to match the simulated white noise in the hardware setup. Although we were able to generate noise using a wave generator in the lab, the noise characteristics did not perfectly match the simulated noise in LTSpice, making it difficult to fully replicate real-world noise conditions during hardware testing.

Solution Implemented:

To overcome these challenges, we used a voltage divider network to generate the missing voltage source, ensuring stable reference voltage. For noise simulation, although the generated noise in the lab differed slightly from the simulated noise, we conducted multiple tests with varying input signals to analyze the circuit's stability under different conditions. Lastly, the recalculation of threshold voltages ensured that the deviation caused by mismatched resistor values did not affect the circuit's overall performance. These solutions helped maintain the circuit's desired functionality and ensured that the project objectives were successfully achieved.

9. Conclusion:

The project successfully implemented an Inverting Schmitt Trigger to create a hysteresis window between 1V and 3V, helping to stabilize noisy input signals. The circuit worked as expected, with the output switching correctly between HIGH and LOW when the input crossed the threshold voltages. Some challenges were faced, such as the lack of a third voltage source and slight differences in resistor values, but these were solved by using a voltage divider and adjusting the threshold values. Multiple tests with different input signals confirmed that the circuit was stable and performed reliably. This project helped us understand how to design and build circuits while also learning how to solve practical issues during hardware implementation. Overall, the project met its objectives and provided useful learning experiences.

References

- [1] Sedra A and Smith K C, *Microelectronic circuits*, 6th ed. New York: Oxford University Press, 2010.
- [2] Cadence PCB Solutions, "Schmitt Trigger Hysteresis Provides Noise-Free Switching and Output," *Cadence Blog*, Sep. 14, 2021. [Online]. Available: <https://resources.pcb.cadence.com/blog/2021-schmitt-trigger-hysteresis-provides-noise-free-switching-and-output>
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